

Impacket Security Assessment

TM-022 Purple Validation Harness · Wave 1

Project	iHBV-TM-022
Author	HoneyBadger Vanguard LLC
Date	2026-08-14
Classification	Internal — Purple Team Research

Impacket Security Assessment

Program: TM-022 Purple Validation Harness — Wave 1 **Project code:** iHBV-TM-022 **Subject:** `fortra/`
`impacket` (Kerberos subsystem + supply-chain surface) **Assessor:** HoneyBadger Vanguard LLC **Date:** 14
August 2026 **Distribution:** Internal — Purple Team Research

Executive Summary

This assessment covers two attack surfaces of the `impacket` library as consumed on Windows-domain engagements:

1. **Correctness of the Kerberos credential-cache (ccache) parser** — the code that decides which cached ticket is used for a given request. A merged fix (PR #2242, master head `4c09897`) addresses one known crash class; this engagement adversarially tested six spec-derived boundary conditions across the pre-fix baseline (`243d64a^`, 0.13.1-equivalent) and the patched master.
2. **The dependency supply-chain** — every third-party Python package that ships alongside `impacket`, statically resolved and analysed with the `depSNORT` tool.

Headline findings

ID	Class	v0.13.1	Master	Priority
EC-04	UTF-8 principal handling	crash + silent mojibake	unchanged	HIGH
EC-06	Expired / bad-flag ticket selection	silent-wrong-result	unchanged	HIGH
EC-R	No- / search SPN with anySPN=True	crash	crash (RESIDUAL)	Medium
EC-03	S4U2Self ST for a user principal	silent-skip	unchanged	Medium
EC-01	3-part SPN in anySPN loop	crash (poisons whole cache)	fixed	—
EC-05	Port specifier in SPN 2nd component	3-part crash	fixed	—
EC-02	Cross-realm referral TGT	correct	correct	—
DEP-01	Unpinned requirements.txt	—	gate-eligible (exit 3)	Medium
DEP-02	pyasn1 dormancy signal (VC-004)	—	advisory	Watch
DEP-03	Install-hook capabilities	—	clean (0/10 hooks)	—

Two findings the vendor's fix does not touch (EC-04, EC-06) outrank EC-01 on operational risk, per the "silent-wrong-result outranks crash" heuristic governing this engagement. A residual crash of the same structural class as EC-01 survives on master (EC-R) — the fix hardened the cached-server side of the SPN parser but left the search-server side exposed. The supply-chain scan returned zero block-class findings against the current transitive tree; the one gate-eligible finding is that requirements.txt is entirely unpinned, which prevents any CI supply-chain scanner from acting on a stable graph.

1. Scope and Objectives

In scope

- `impacket/krb5/ccache.py` — Kerberos credential-cache parser and the `CCache.getCredential` service-principal matcher.
- Cross-file structural-assumption sweep across `impacket/krb5/` for the patterns identified in the primary target file.
- `impacket`'s declared and transitive Python dependency graph via `setup.py` and `requirements.txt`.

Out of scope

- Live directory / Kerberos infrastructure. All fixtures are constructed in-memory or from crafted lockfiles; no live AD, KDC, or on-wire traffic.
- The SMB, LDAP, and DCE/RPC subsystems, beyond passing observation of the ccache API's callers.
- Rubeus and bloodyAD (assigned to separate Wave 1 sessions).

- Detection-side artefacts (a separate workstream).

Objectives

1. Confirm the vendor's PR #2242 fully resolves the seed edge case (EC-01) under all spec-legal permutations, not merely the one in the original bug report.
2. Characterise five additional spec-derived boundary conditions (EC-02–EC-06) that a flat single-domain test lab would not produce.
3. Identify any residual instances of the same structural assumption elsewhere in the code base.
4. Produce an independent supply-chain risk view of `impacket`'s dependencies.

2. Methodology

2.1 Kerberos ccache (EC-01 – EC-06)

- **Spec first, code second.** Each edge case was scoped from RFC 4120, MS-KILE, or MS-SFU before the implementation was consulted, to prevent the implementation from anchoring expectations of what "should" work.
- **Fixture-driven and self-contained.** ccache principals were constructed in-memory via `types.Principal(<spn>) → ccache.Principal.fromPrincipal`, using the same API path that `fromTGT` and `loadKirbiFile` exercise. No live AD, no raw byte crafting. Fixtures are regenerable against future versions.
- **Differential across two versions.** All fixtures were run against both the pre-fix baseline (`243d64a^`, whose `getCredential` is byte-identical to the 0.13.1 source cited in the seed report) and the patched master (`4c09897`), with strict module isolation via `PYTHONPATH` from a neutral working directory.
- **Silent-wrong-result outranks crash.** A crash is loud and fails fast; a mis-selected credential produces confusing wire-level failures or, worse, uses a ticket the operator did not intend. Silent-wrong-result findings are prioritised above `IndexError` crashes.

2.2 Supply chain (DEP-01 – DEP-05)

- Static, non-executing scan via `depSNORT` (build `dev`, commit `5b02377`, Go 1.24, `CGO_ENABLED=0`). `depSNORT` parses lockfiles and manifests only; it never invokes `pip` or fires an install hook.
- Two runs against `impacket`: one against the repository as-shipped (unpinned `requirements.txt`), one against a machine-generated pin set produced by installing the declared deps into a scratch environment and reading `importlib.metadata`.
- Multiple output formats emitted (JSON, SARIF, DOT, PDF) for downstream ingestion.

2.3 Version identifiers

Component	Reference	Commit
impacket baseline (0.13.1-equivalent)	243d64a^	2dae5c7
impacket master (patched)	master	4c09897
Vendor fix (PR #2242)	243d64a	—
depSNORT	master	5b02377

The `impacket_0_13_1` tag is not present in the `MoSLoF/impacket` fork, so the baseline was taken as the commit immediately preceding the fix; its `getCredential` implementation is line-for-line identical to the 0.13.1 source quoted in the seed bug report and is therefore a faithful proxy for the function under test.

3. Environment

- Container: Linux 6.18.5 sandbox, Python 3.11, Go 1.24.7.
 - Outbound egress is mediated by an agent proxy; `api.osv.dev` is denied at the gateway (HTTP 403 CONNECT). This affected only the OSV advisory tier of the supply-chain scan and is disclosed in the DEP-04 finding. All temporal registry sources (PyPI, npm, RubyGems, Cargo, Composer, NuGet) were reachable and cached locally.
-

4. Findings — Kerberos ccache

4.1 Summary matrix

EC	Condition	v0.13.1	Master	Same assumption elsewhere	Priority
EC-01	3-part SPN in <code>anySPN</code> loop	crash (poisons cache)	correct	search-side residual (EC-R)	— (fixed)
EC-02	Cross-realm referral TGT	correct	correct	n/a	Low
EC-03	S4U2Self ST, no service type	machine ok; user silent-skip	same	n/a	Medium
EC-04	Non-ASCII (UTF-8) principals	crash + mojibake	unchanged	systemic <code>six.b()</code> latin-1	High
EC-05	Port specifier in 2nd SPN component	3-part: crash	correct	(subsumed by EC-01 fix)	— (fixed)
EC-06	Expired / bad-flag ticket	silent-wrong-result	unchanged	selection layer ignores time & flags	High
EC-R	No- / search SPN	crash	crash (residual)	same file, search-side	Medium

4.2 EC-01 — 3-part SPN in `CCache.getCredential` `anySPN` fallback

- **Spec basis:** MS-KILE §2.2 / RFC 4120 §6.2. An SPN's `serviceName` is a sequence of `KerberosString` components; three-component service principals (`service/host/domain`) are spec-legal and are produced when Windows caches a service ticket dumped from LSASS memory (e.g. via `Rubeus dump`), with the domain suffix appended as a third component.
- **File:** `impacket/krb5/ccache.py` :: `CCache.getCredential` (`anySPN` loop).
- **Baseline behaviour:** the `if find(b'/' ) >= 0` branch performs `split(b'/' )[1].split(b'@' )[1]` . For `LDAP/host/domain@REALM` , `split(b'/' )[1]` returns the *host* component, which contains no `@` , so `.split(b'@' )[1]` raises `IndexError` .
- **Master behaviour:** the new `count(b'/' ) >= 2` branch (`ccache.py`:431–454) parses 3-part SPNs, strips an optional port from the 2nd component, handles ≥ 4 -part shapes correctly, does short-name $\rightleftharpoons$ FQDN matching only when at least one side is unqualified, and requires an exact realm match. Verified against all spec-legal permutations enumerated in the fixture harness.
- **Operational severity of baseline (poisoning):** the `anySPN` loop iterates every credential in the cache; a single 3-part ticket anywhere in `KRB5CCNAME` raises before the loop can reach the credential the caller asked for. Because `CCache.parseFile` unconditionally calls `getCredential('krbtgt/DOMAIN@DOMAIN' , anySPN=True)` (`ccache.py`:667), a single dumped LDAP ticket in the operator's cache breaks plain TGT retrieval for the entire toolchain — `secretsdump` , `smbclient` , `getST` , and every other consumer — not merely LDAP operations.
- **Verdict:** the vendor fix is correct and complete for the cached-server side of `getCredential` . See EC-R for the residual search-side gap.

4.3 EC-02 — Cross-realm referral TGT `krbtgt/child.realm@parent.realm`

- **Spec basis:** RFC 4120 §3.3.3.
- **Result:** correct on both versions. A referral TGT is a well-formed two-part SPN carrying `@realm`, so it takes the standard branch and does not trip the EC-01 crash. It does not poison following credentials.
- **Priority:** Low. A latent, low-consequence note is that the two-part branch treats the referral's `CHILD.REALM` as the "host" component; a search for a service on a host literally named `child.realm` in `parent.realm` could in principle collide. This is not a realistic operational input.

4.4 EC-03 — S4U2Self ST with no service type (server principal has no `/`)

- **Spec basis:** MS-SFU §3.2.5.1.2. An S4U2Self ST's `sname` is the requesting principal; it may be a user (no service class) or a machine account (`host$`).
- **File:** `impacket/krb5/ccache.py :: getCredential`, no-slash branch.
- **Result:** the branch hardcodes `hostname$@REALM ( searchSPN = "<short>${<realm>}" )`.
 - Machine accounts match correctly (intended path).
 - **User principals silently miss:** an S4U2Self ST for `Administrator@REALM` (no `$`) is never matched by the `anySPN` fallback, because the appended `$` guarantees a miss.
- **Severity:** Medium. The failure is fail-closed (returns `None`, caller falls back to requesting a fresh ticket), so it does not corrupt state; however, an operator relying on a cached user-ST via `anySPN` will silently not get one. Present identically on both versions.

4.5 EC-04 — Non-ASCII (UTF-8) principal / realm handling (HIGH)

- **Spec basis:** RFC 4120 §5.2.1 (KerberosString = IA5String, ASCII) extended by MS-KILE and real-world Windows/MIT/Heimdal practice to UTF-8 principal and realm names.
- **File:** `impacket/krb5/ccache.py :: Principal.prettyPrint, getCredential`. Root cause is systemic use of `six.b()` (latin-1 encoding) across the code base; commit `8ea54fe` ("Modernize byte string normalization", PR #2243) modernised `dcerpc`, `ntlm`, and examples but does not touch `ccache.py` or the principal-encoding path.
- **Two distinct failure modes:**
 1. **Silent mojibake (latin-1 range: `é ü ö ä ñ ...`). `six.b('café')` returns `b'caf\xe9'` (latin-1), not the UTF-8 `b'caf\xc3\xa9'` that real ccaches store. When a `Principal` is built from a string, its components are stored as `str` and re-encoded latin-1 on `prettyPrint / getData`. A ccache **written** by `impacket` with accented names is therefore byte-incompatible with MIT, Heimdal, and Windows, and cross-tool SPN comparison silently mismatches.**
 2. **Hard crash (outside latin-1: Cyrillic, Greek, CJK, emoji, Hebrew). `six.b('MOCKBA')` raises `UnicodeEncodeError`. This fires on the very first line of `getCredential ( b(server.upper()) )`, so any lookup whose search SPN contains a non-latin-1 character crashes; the same is true for `parseFile` on a cache whose principal has such a name.**
- **Read-only display of a genuine UTF-8 ccache is safe** — bytes loaded from disk pass through `prettyPrint` untouched. The break is on the construction, comparison, and re-encoding paths.
- **Operational impact:** engagements in organisations that use accented or localised `SAMAccountName` values or realm strings (common in EMEA, LATAM, and APAC) encounter silent wrong-results or hard stops with no obvious cause. Identical behaviour on both versions.

4.6 EC-05 — Port specifier in SPN second component

- **Spec basis:** MS-KILE SPN grammar: `serviceclass "/" hostname [":" port] ["/" servicename]`.
- **Result:**
 - Two-part with port (`HTTP/web.lab.local:8443`): both versions strip the port and match correctly.
 - Three-part with port (`LDAP/dc.lab.local:3268/lab.local@REALM`) against a two-part search: **baseline** `IndexError` — a second independent trigger of the EC-01 crash class. Master strips the port in the new 3-part branch and matches correctly.
- **Verdict:** fully resolved on master. All combinations verified (two-part, two-part+port, three-part, three-part+port, four-part).

4.7 EC-06 — Mixed / expired / unusual ticket flags (HIGH)

- **Spec basis:** RFC 4120 §5.3 (`endtime` , `renew-till`) and §2 (`TicketFlags` : FORWARDABLE, RENEWABLE, and related bits). A consumer choosing a cached ticket is expected to honour the validity window and any required flags.
- **File:** `impacket/krb5/ccache.py` :: `getCredential` (selection layer).
- **Result:** `Credential` parses and stores `time.endtime` , `time.renew_till` , and `tktfldgs` , but these fields are consulted only for pretty-printing and re-serialisation. `getCredential` selects the **first structural SPN match**, unconditionally.
- **Verified:** a cache holding, for the same SPN, an expired non-forwardable ticket at index 0 and a valid forwardable ticket at index 1 returns the expired non-forwardable one. Order-dependent; identical on both versions.
- **Operational consequences:**
 - *Expired ticket:* no local pre-flight check; the dead ticket is handed downstream and fails at the KDC/target (`KRB_AP_ERR_TKT_EXPIRED`) — a confusing wire-level failure rather than a clean "ticket expired, re-authenticating" locally.
 - *Flags:* a delegation flow (S4U2Proxy) that requires FORWARDABLE will happily receive a non-forwardable cached TGT; the requirement is only ever (perhaps) enforced by the KDC, not by Impacket's selection. "Forwarded but not forwardable" and non-renewable tickets are likewise never filtered.

4.8 EC-R — Residual: no- / search SPN with `anySPN=True` (survives on master)

- **Class:** same as EC-01 — an unconditional `split('/')[1]` index on the *search* side.
- **File:** `impacket/krb5/ccache.py` :: `getCredential` , lines 458–459 and 466 (search side).
- **Result:** the EC-01 patch hardened the *cached-server* side against missing components but left the *search-server* side assuming the caller's SPN always contains a `/` (`server.upper().split('/')[1]` , `ccache.py`:458, 459, 466). If `getCredential` is called with a no-slash target (e.g. a bare `host@REALM` or `user@REALM`) **and** the cache contains any slash-bearing credential, the `anySPN` loop raises `IndexError` on master.

- **Reachability:** `CCache.parseFile` builds `principal = '<target>@<domain>'` from a caller-supplied `target` (`ccache.py:660`). Every in-tree caller currently passes a full SPN or `krbtgt/...` (both contain `/`), so the primary tools are not exposed on the current call sites. `getCredential` is public API, however, and is used by external tooling and by `examples/getST.py:184` with the default `anySPN=True`; any caller passing a no-slash target trips it.
- **Recommendation:** apply the same defensive guard (`len(parts) > 1 / count('/', '')`) on the search side. This is a one-line fix of the same class as the vendor's cached-side patch.

4.9 Cross-file assumption sweep

`grep -rn "split(b'/'\)[1\]|split('/'\'[1\]|split('@'\'[1\]"` across `impacket/krb5/` returns matches only in `ccache.py :: getCredential` (lines 457–466). The unconditional-index SPN pattern is not duplicated in `kerberosv5.py`, `kpasswd.py`, `types.py`, or elsewhere. `kpasswd.py:363` invokes `getCredential(principal, anySPN=False)`, which never enters the fragile loop.

The `six.b()` latin-1 assumption underlying EC-04, by contrast, is **systemic** across the code base. Commit `8ea54fe` modernised several modules but explicitly did not touch `ccache.py` or the principal-encoding path.

5. Findings — Supply Chain

5.1 Summary matrix

ID	Class	Gate	Impact
DEP-01	Unpinned deps (D-01)	gate-eligible (exit 3)	Reproducibility gap; scanners cannot gate on a fixed graph
DEP-02	VC-004 dormancy on <code>pyasn1@0.6.4</code>	Advisory	Monitor next <code>pyasn1</code> release; crypto-critical dep
DEP-03	VC-002 install-hook family	Clean (0/10 capable)	No exfil/cradle install hooks in transitive tree
DEP-04	OSV.dev 403-blocked by environment	Gate-eligible (env)	Sandbox limitation; not an impacket issue
DEP-05	VC-007 dependency confusion	n/a	Not exercised (no internal scopes declared)

5.2 DEP-01 — `requirements.txt` is entirely unpinned

Every one of the ten declared runtime dependencies uses either a floating name (`setuptools`, `six`, `charset_normalizer`, `pyasn1_modules`, `pycryptodomex`, `pyOpenSSL`) or a lower-bound-only specifier (`pyasn1>=0.2.3`, `flask>=1.0`, `ldapdomaindump>=0.9.0`, and the exclusion list on `ldap3>=2.5,!2.5.2,!2.5.0,!2.6`). No upper bound is set anywhere.

Consequences:

- Every fresh `pip install impacket` picks whatever the resolver currently prefers. There is no cross-machine or cross-CI reproducibility of the dep tree.
- A compromised newer minor release of any dep is picked up silently until it is yanked upstream.
- Automated supply-chain scanners (depSNORT, `pip-audit`, `safety`, etc.) cannot act on a stable `package@version` graph. depSNORT correctly refuses to speculatively resolve unpinned specifiers (design decision D-01) and returns exit 3 under `-fail-on-incomplete`.

Recommendation: add a `requirements.lock` produced by `pip-compile` or adopt `uv` / `Pipfile.lock`. Keep `requirements.txt` as the human-authored input; use the lockfile for CI and reproducible installs.

5.3 DEP-02 — `pyasn1@0.6.4` published after 492 days of dormancy (VC-004)

The only substantive finding across the pinned scan. `pyasn1` was silent for 492 days and then released 0.6.4 — the account-takeover shape depSNORT watches for. Per VC-004's design this remains advisory unless the awakening release also declares an install hook, which `pyasn1`'s does not (verified in the install-surface subgraph — no `install-hook` node for `pyasn1@0.6.4`).

No immediate action required, but `pyasn1` is a load-bearing Kerberos and ASN.1 crypto dependency for `impacket`: a compromise of that maintainer account would poison every `impacket` install. **Treat `pyasn1`'s next release as one to watch, not one to auto-upgrade.**

5.4 DEP-03 — Install-surface subgraph: clean

depSNORT statically extracted install-time hooks from every resolved dependency (npm-style lifecycle hooks, Python `setup.py` / `build-backend` / `.pth` files, and equivalents in other ecosystems). All ten `install-hook` nodes carry `capabilities: none` — no network reach, no named-credential access, no decode-and-execute indirection, no download cradle.

Package	Hook site(s)
<code>charset-normalizer 3.4.6</code>	<code>pyproject.toml</code> <code>build-backend</code> , <code>setup.py</code> <code>module-level</code>
<code>ldap3 2.9.1</code>	<code>setup.py</code> <code>module-level</code>
<code>pycryptodomex 3.23.0</code>	<code>setup.py</code> <code>module-level</code> , <code>cmdclass.build_ext</code>
<code>pyOpenSSL 26.4.0</code>	<code>pyproject.toml</code> <code>build-backend</code> , <code>setup.py</code> <code>module-level</code>
<code>setuptools 68.1.2</code>	<code>setup.py</code> <code>module-level</code> , <code>cmdclass.install</code>
<code>six 1.16.0</code>	<code>setup.py</code> <code>module-level</code>

Notably, `setuptools`'s `cmdclass.install` was not misclassified as exfil-capable; depSNORT's precision-first design (VC-002 only flags *named* credential access, not blanket environment reads) held up on the legitimate native-build shape. `impacket`'s own `setup.py` also parsed clean — the `git` sub-process it invokes for versioning is neither a network reach nor a credential read.

5.5 DEP-04 — Coverage: OSV.dev advisory tier unreachable in this environment

The engagement runner's egress proxy denies `api.osv.dev` at the gateway (HTTP 403 CONNECT, verified in `__agentproxy/status`'s `recentRelayFailures`). `depSNORT` reported the degradation correctly and returned exit 3 under `-fail-on-incomplete`. This is a sandbox limitation, not an impacket property, and is documented for reproducibility. On a network-connected host (or with a pre-fetched `-osv-snapshot` file), the OSV advisory tier would run against the same pin set.

5.6 DEP-05 — VC-007 dependency confusion (not exercised)

The dependency-confusion check requires the operator to declare internal scopes/names via `-internal-scopes` / `-internal-names`. `impacket` ships no internal package names, so VC-007 correctly no-op'd. Recorded here so the matrix is complete.

6. Recommendations

Ranked by operational impact.

1. **EC-04 (High) — normalise principal handling to UTF-8.** Replace `six.b()` on the principal-encoding path with explicit `str.encode('utf-8')` on write and `bytes.decode('utf-8')` on compare. Add fixtures with accented and non-latin-1 principal names to `tests/misc/test_ccache.py`. This closes both the silent-mojibake and the hard-crash paths.
 2. **EC-06 (High) — filter cached tickets on validity and required flags.** The selection layer (or its callers) should skip expired tickets and accept a `required_flags` argument, or at minimum expose `endtime` and `tktfags` on the return value so consumers can filter. First-match selection over an unfiltered list is the wrong default for a ticket cache.
 3. **EC-R (Medium) — apply the same length guard on the search side of `getCredential`.** One-line defensive check (`if '/' in server: / len(parts) > 1`) at `ccache.py:458` and `466`. This closes the residual crash of the class that PR #2242 was opened to address.
 4. **EC-03 (Medium) — allow the no-slash branch to match user principals**, or document that `anySPN` resolves only machine-account S4U2Self STs.
 5. **DEP-01 (Medium) — add a machine-generated pin set** so CI can gate on a stable graph, supply-chain scanning has a concrete `package@version` to match against, and dormancy-then-release events (DEP-02) become actionable rather than advisory noise.
-

7. Reproducibility

All fixtures and scans are committed on branch `claude/impacket-adversarial-wave-1-39koer`.

Kerberos ccache differential

```
# baseline (0.13.1-equivalent)
git worktree add /tmp/prefix-tree 243d64a^

# from a NEUTRAL working directory (repo root shadows PYTHONPATH)
(cd /tmp && PYTHONPATH=/tmp/prefix-tree python3 tm022_wave1_harness/harness.py)
(cd /tmp && PYTHONPATH=/path/to/impacket python3 tm022_wave1_harness/harness.py)

# EC-04 layer analysis
(cd /tmp && PYTHONPATH=... python3 tm022_wave1_harness/ec04.py)
# EC-06 ordering test
(cd /tmp && PYTHONPATH=... python3 tm022_wave1_harness/ec06.py)
```

Supply-chain scan

```
# depSNORT build (Go 1.24+, no network needed)
CGO_ENABLED=0 go build -o depsnort ./cmd/depsnort

# repo scan (unpinned)
./depsnort scan /path/to/impacket

# CI gate on unpinned specifiers
./depsnort scan -fail-on-incomplete /path/to/impacket          # exit 3

# pinned scan (using the committed pin set)
./depsnort scan tm022_wave1_harness/depsnort-pinned-requirements.txt
```

Appendix A — Committed artefacts (branch `claude/impacket-adversarial-wave-1-39koer`)

Path	Contents
<code>TM-022_Wave1_impacket_findings.md</code>	Full Kerberos ccache findings (this report's §4)
<code>TM-022_Wave1_depSNORT_findings.md</code>	Full supply-chain findings (this report's §5)
<code>TM-022_Wave1_depSNORT_report.pdf</code>	depSNORT-generated PDF of the pinned scan
<code>tm022_wave1_harness/harness.py</code>	Differential harness — EC-01, EC-02, EC-03, EC-05, and pathological shapes
<code>tm022_wave1_harness/ec04.py</code>	UTF-8 principal layer analysis
<code>tm022_wave1_harness/ec06.py</code>	Expired-ticket / flag ordering test
<code>tm022_wave1_harness/depsnort-scan-repo.json</code>	depSNORT JSON — unpinned scan
<code>tm022_wave1_harness/depsnort-scan-pinned.json</code>	depSNORT JSON — pinned scan
<code>tm022_wave1_harness/depsnort-scan-pinned.sarif</code>	depSNORT SARIF — CI ingestion
<code>tm022_wave1_harness/depsnort-pinned-requirements.txt</code>	Reproducible 10-of-10 pin set
<code>reports/01_impacket_security_assessment.pdf</code>	This report, PDF form
<code>reports/02_depSNORT_tool_assessment.pdf</code>	Separate tool evaluation
<code>reports/03_session_transcript.pdf</code>	Complete assessment transcript

Appendix B — Version identifiers

Component	Reference	Commit
impacket baseline	<code>243d64a^</code> (0.13.1-equivalent)	<code>2dae5c7</code>
impacket master	<code>master</code> (patched)	<code>4c09897</code>
Vendor fix (PR #2242)	<code>243d64a</code>	—
depSNORT	<code>master</code>	<code>5b02377</code>

This report is internal HoneyBadger Vanguard LLC research under project iHBV-TM-022 (TM-022 Purple Validation Harness). Distribute only within the purple-team programme.